Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and AI systems rarely co-train encoder and memory across a graded sensor design. We test this in silicon: a navigation agent where only the visual sensor varies along a graded scale of encoder spatial bandwidth, with architecture, task, and training held fixed. Across five conditions, agents reach near-identical task performance yet the hidden-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the hidden state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift, not a total-information collapse. We frame this as an emergent *capacity allocation* between the encoder route and the memory-integration route, measured along three concomitant aspects (magnitude, format, consumption). Memory transplants are asymmetric (rich-encoder policies fail on bottleneck states; the reverse works), paralleling cross-modality remapping in hippocampal coding under hidden-state inference. Probe-readable position further dissociates from policy reliance: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. This positions sensor structure as a training-time lever on cognitive-map format, generates a falsifiable architecture-agnostic prediction for transformer navigators, and reframes bandwidth-restricting perception as potentially cognition-enabling rather than only compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a sequential pipeline, guided by the implicit assumption that scaling the encoder uniformly improves downstream representations. Recent benchmarks complicate this picture: vision–language models with strong visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025]. Existing explanations have focused on the training distribution [Chen et al., 2024] or on differences in the geometric content of pretrained encoders [Tong et al., 2024, El Banani et al., 2024]. We instead question the underlying assumption: by treating perception and memory as a co-trained system, we ask how the structure of the encoder constrains what spatial information is accessible from memory through simple linear readout.

Cognitive neuroscience has long recognised that sensor structure shapes the format of cognitive maps. Across species, distinct sensory ecologies drive distinct spatial-coding strategies — foveated primates rely on view cells while near-panoramic rodents rely on place cells [Wirth et al., 2017, Rolls, 2024]; echolocating bats and congenitally blind humans show striking cross-modal remapping [Geva-Sagiv et al., 2016, Fortin et al., 2008]. Empirically isolating the causal role of sensor ecology is notoriously difficult in biology: body plans and evolutionary histories cannot be held

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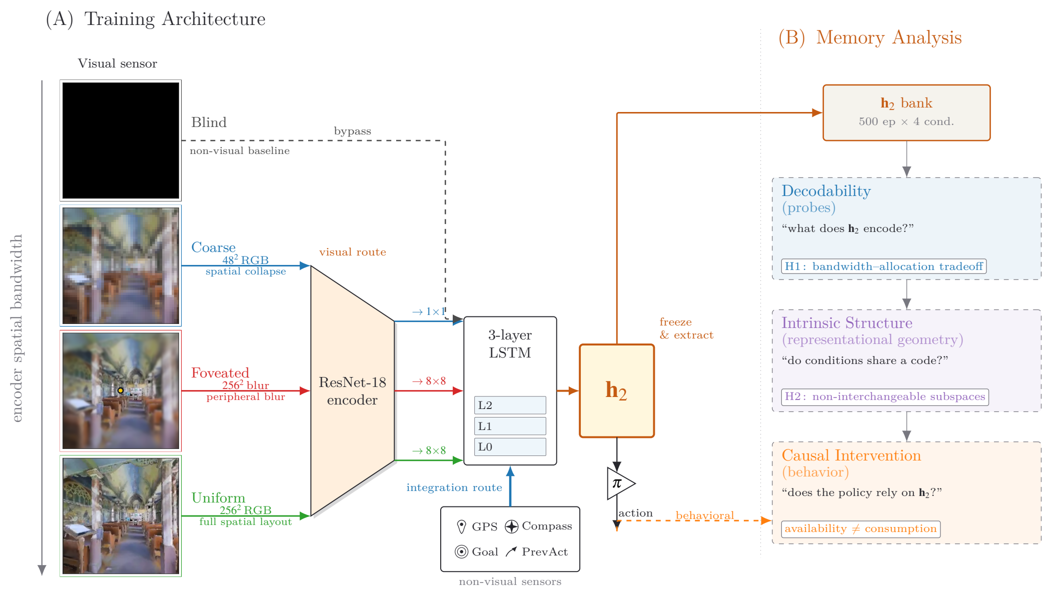


Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span a graded scale of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix I) adds a 5th intermediate point on the same scale. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three views of one capacity allocation: *magnitude*, *format*, *consumption* (ğ2; synthesis in ğ4).

constant, so cross-species effects are confounded with everything else that differs between species. Artificial systems offer a powerful alternative, yet modern deep reinforcement learning rarely co-trains perception and memory across a systematically controlled sensor design.

We bridge this gap *in a silicon setting*. Building on deep-RL navigation agents known to develop cognitive-map-like representations even in the absence of vision [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018],[1] we introduce a comparative testbed that holds the encoder backbone, recurrent memory, task, optimisation, and reward entirely fixed, and varies a single degree of freedom: the spatial bandwidth of the visual sensor across five conditions (Figure 1).

We hypothesise an emergent *capacity-allocation* principle: the visual encoder and the recurrent integration of continuous sensor history (e.g., GPS) compete for a finite hidden-state budget. Our results confirm this tradeoff. As encoder spatial bandwidth increases, the linearly-readable, world-frame position signal in the recurrent state collapses monotonically while task performance stays near-perfect (93–99% success). We establish this reallocation along three concomitant views: *magnitude* (how much linearly-accessible position information lives in the recurrent state), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). Robustness against single-method artefacts is confirmed by convergent results across linear, non-linear (MLP), and information-theoretic (MINE) probes, leave-one-scene-out validation, and Procrustes shape distance. We further ground the format shift behaviourally with asymmetric memory transplants: rich-encoder policies fail to navigate using bottleneck states, demonstrating the representational change is mechanistically intrinsic and not an artefact of the probing framework [Orozco et al., 2025].

[1]"Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

Our contributions are three folds. **(i) The capacity-allocation hypothesis:** a unifying framework explaining how visual bandwidth dictates spatial-memory format, organising five concomitant signatures (magnitude collapse, format divergence, consumption dissociation, scene-invariance, place-vs-view-cell character) into three concomitant views (magnitude, format, consumption). This matched-condition design enables an analytical avenue biology cannot easily realise: the systems-neuroscience toolkit applied to a *controlled comparative population* rather than a single recording. We treat this comparative methodology as an explicit secondary contribution, developed in full in ğ5. **(ii) Causal behavioural validation:** asymmetric memory transplants demonstrating that rich-encoder policies fail on bottleneck states, mechanically grounding the format shift and mirroring biological global remapping under cross-modality reorganisation. **(iii) Broader theoretical implications:** a falsifiable prediction for transformer-based navigators (ğ4) and an architecture-side reframing of bandwidth-restricting perception (foveation, log-polar) as cognition-*enabling* rather than only compute-saving — complementing data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures.

2 Methods

Architecture, task, and conditions (Figure 1, Row 1). PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020] across five conditions varying only the visual sensor: *blind* (no encoder); *coarse* ($48{\times}48$ RGB, $1{\times}1$ encoder feature map after ResNet-18 downsampling); *foveated* ($256{\times}256$ with eccentricity-dependent Gaussian blur $\sigma_{\max}$=8, $4{\times}4$ feature map); *uniform* ($256{\times}256$ unblurred, $4{\times}4$); *foveated-logpolar* ($\sim 2{\times}2$ via log-polar resampling onto a $64{\times}64$ retinal grid, control point isolating encoder-spatial-output from blur, Appendix I). All five share an identical 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal); Gibson-0+ [Xia et al., 2018] and MP3D training pool; 18 MP3D test scenes for distribution-shift probes; 250M frames each; single seed per condition (cogsci modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]; rigour from convergent multi-method evidence, ğ4.4); blind has an independently re-trained checkpoint (Appendix A). After training, weights are frozen and per-step top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ is collected from $n = 500$ Gibson held-out deterministic-rollout episodes per condition.

Probing methodology (Figure 1, Row 2). Three method blocks read $\mathbf{h}_2$ to test three concomitant aspects of one capacity allocation:

- *Decodability $\rightarrow$ magnitude*: linear Ridge with Hewitt–Liang selectivity [Hewitt and Liang, 2019], 2-layer MLP, MINE [Belghazi et al., 2018], Skaggs spatial information [Skaggs et al., 1996]; plus step-binned, lag-k, predictive-horizon [Stachenfeld et al., 2017, Whittington et al., 2020], and excursion-forgetting variants for memory dynamics.
- *Intrinsic structure $\rightarrow$ format*: LOSO cross-validation [Sanders et al., 2020], cross-condition probe transfer, CKA [Kornblith et al., 2019], 1-NN purity, Procrustes shape distance [Williams et al., 2021], principal angles and position-direction cosines, participation ratio and TwoNN intrinsic dimension [Ansuini et al., 2019, Facco et al., 2017], eigenspectrum power-law [Stringer et al., 2019].
- *Causal intervention $\rightarrow$ consumption*: $5{\times}5$ memory transplant and memory-init transplant; paired same-scene shortcut-discovery (Tolman 1948 in silico); GPS-sensor mid-rollout ablation.

Each method is introduced at point of use in ğ3; full protocols and hyperparameters in Appendix B. Convergence across the three views on the same condition-level contrast is the strongest claim the methodology supports.

3 Results

The five agents reach near-identical task performance (Table 1) despite substantially different visual inputs — the encoder–memory–policy stacks must therefore internally compensate for the input differences, holding different internal representations of the same task. We read those representations along four views matching the three method blocks of Figure 1: how much linearly-readable posi-

Table 1: Per-condition headline. All five agents trained for 250M frames (ğ2); blind uses an independently retrained checkpoint (Appendix A). Probe R^2: 5-fold episode-level CV mean ± std on full-length trajectories (no per-episode step cap). Sub-chance R^2 in rich-encoder rows reflect long-tail collapse (mid-episode $R^2 \in [0.6, 0.95]$, Figure 9b); rich-encoder GPS R^2 is statistically indistinguishable from zero (one-sample t-test: foveated $p \approx 0.58$, uniform $p \approx 0.25$; df=4). Bold marks blind/coarse rows where GPS and compass remain decodable ($p < 0.001$) and best task performance.

Condition	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	0.853	0.989	$+\mathbf{0.582 \pm 0.099}$	$+\mathbf{0.659 \pm 0.061}$
Foveated	**0.894**	**0.993**	$+0.178 \pm 0.659$	$+0.503 \pm 0.142$
Foveated-logpolar	0.879	0.990	-0.029 ± 0.759	$+0.467 \pm 0.330$
Uniform	0.891	**0.993**	-1.785 ± 2.993	-1.244 ± 3.579

tion lives in memory (magnitude, ğ3.1); what shape it takes spatially (ğ3.2) and temporally (ğ3.3); and what behaviour reads from memory under causal intervention (consumption, ğ3.4). Each view points to the same underlying capacity-allocation tradeoff between an encoder-derived visual route and a memory-integration route. The Discussion (ğ4) applies the same set of analyses to a different environment and encoder family, as a step toward a comparative cognitive neuroscience of artificial navigation agents.

3.1 Magnitude: linear position code shrinks with encoder bandwidth

Memory has two routes to a position code: integrate motion over time (works even without vision) or read position-correlated visual features (only works with rich vision). The policy reads memory through a linear operation, so capacity allocation between the two routes is set by which one the policy can read. As we widen the visual encoder, the agent reallocates from the first route to the second, and **the linearly-readable position signal at the policy-readable layer shrinks monotonically with encoder bandwidth** (Figure 2a). All five agents reach near-ceiling task performance (Table 1); this is a representational dissociation, not a skill gap. The position information itself is not lost — a non-linear probe recovers it (ğ3.2) — only the linear, policy-accessible component shrinks.

Two natural alternative readings are ruled out by controls, and a sharper picture of the trigger emerges along the way. The chance-level rich-encoder GPS R^2 is a real property of the hidden state, not a probe limitation: the same linear decoder reads distance-to-goal at $R^2 \geq 0.81$ in every condition (Appendix C, Figure 17). The discriminator is also not encoder cell count — and that is the more interesting finding. FOVEATED-LOGPOLAR's encoder produces a $\sim 2{\times}2$ feature map close to coarse's $1{\times}1$, yet lands firmly in the rich-encoder regime, so the relevant variable is per-step *feature variety*, the diversity of visual features delivered at each timestep, not the number of feature cells (Appendix I). The bandwidth axis becomes a sharper, more falsifiable prediction in the process: it should track per-step feature variety in any encoder family, not cell count. Read this way, blind reproduces Wijmans's emergent-cognitive-map result for goal-driven agents without vision [Wijmans et al., 2023], and coarse extends it to a *vision-with-bottleneck* regime — a visual encoder is present, but its $1{\times}1$ output collapses per-step variety to one channel and leaves the integration route to dominate the linear position code.

Capacity allocation across training and predictive horizon. Two further views in Figure 2 read the same allocation differently. Across training (panel b), sighted conditions start with high linear readability; rich-encoder conditions then decay along trajectories whose timescale tracks encoder informativeness, while coarse plateaus and blind stays stable. The successor-representation cognitive-map signature [Stachenfeld et al., 2017] — memory predicting where the agent will be many steps ahead (panel c) — is sustained only in blind, where the integration route carries the position code.

Three appendix diagnostics converge on the same picture (Appendix Figure 9): position readability is comparable across conditions where the position sensor first enters memory and only diverges at the policy-readable layer (a pipeline view); the most spatially-informative units in rich-encoder agents track trajectory-phase and visual-landmark features rather than raw position, while in blind they track raw position — a view-cell-like vs. path-integrated dissociation [Skaggs et al., 1996, Rolls, 2024]; and a 25-step random-action mid-episode excursion fragments blind's position de-

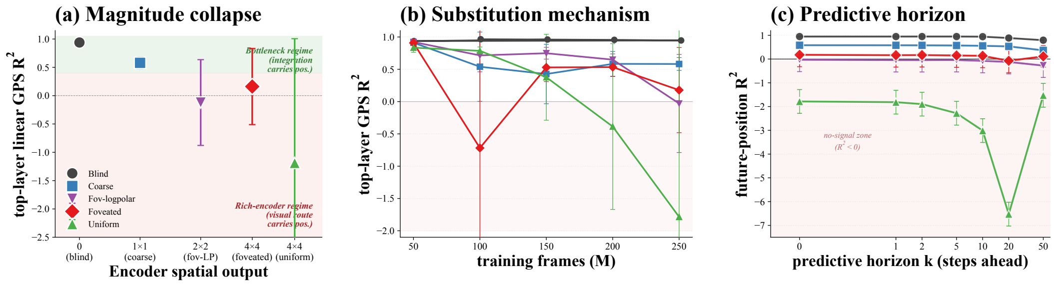


Figure 2: **Three views of magnitude.** *(a) Cross-condition:* top-layer linear GPS R^2 falls with encoder bandwidth. *(b) Across training:* rich-encoder conditions start high then lose readability; bottleneck conditions hold it throughout. *(c) Predictive horizon:* only bottleneck conditions linearly decode pos_{t+k} at long k. Shaded band: $R^2 < 0$, the linear probe is worse than predict-mean baseline — the "no-signal zone". Uniform's curve dives deep into this zone (its hidden state is scene-conditional, so 5-fold CV trains on some scenes' position-directions and tests on incompatible ones, producing systematically negative R^2); the dive is noise, not a meaningful trend.

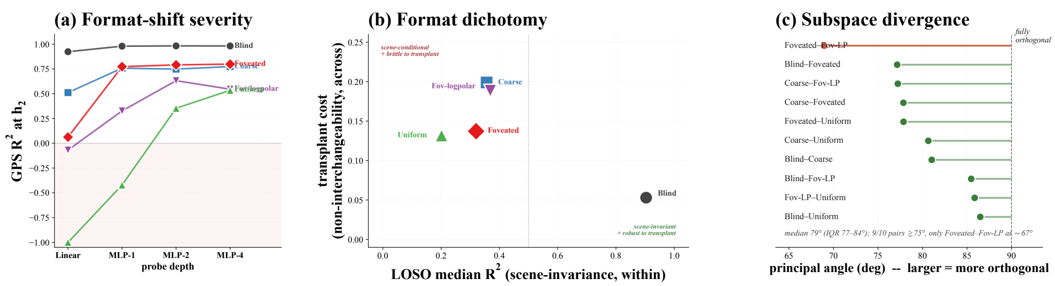


Figure 3: **Three views of spatial format. (a) Linearity (probe-depth sweep):** GPS R^2 vs probe depth (linear $\rightarrow$ MLP-1 $\rightarrow$ MLP-2 $\rightarrow$ MLP-4). Blind is essentially fully linear; coarse intermediate; rich-encoder conditions heavily non-linear, depth required growing with encoder bandwidth. **(b) Scene-context dependence:** LOSO R^2 on x (single-axis transfer across rooms); cross-condition memory-transplant cost on y. Blind sits alone in the scene-invariant + transplant-robust quadrant; the four sighted conditions cluster in the scene-conditional + transplant-brittle quadrant. **(c) Subspace divergence:** principal angle between every pair of conditions' position-direction subspaces. 9/10 pairs at ≥ 75; only the foveated/foveated-logpolar pair at ~ 67.

coding most, consistent at the high level with rodent path-integration under removed visual landmarks [Chen et al., 2016].

3.2 Spatial Format: collapsed linear readability is a format shift, not an information loss

The magnitude collapse of ğ3.1 does not destroy position information: non-linear probes recover position where linear probes fail (Figure 3a), and information-theoretic estimates shift far less than the linear R^2 (Appendix D.1). **The collapse is a format shift, not an information loss.** Format is multi-dimensional — linearity, scene-invariance, subspace identity — and the encoder shifts position along all three, but at boundaries that do not coincide.

To test linearity, we sweep probe depth from a linear classifier to a 4-layer MLP and read where R^2 plateaus (Figure 3a). Blind is fully linear ($R^2 \approx 0.92 \rightarrow 0.97$ across depths). Coarse is intermediate ($0.55 \rightarrow 0.78$). The three rich-encoder conditions are heavily non-linear (linear $R^2 \lesssim 0$, ramping up with depth), with the required depth growing with encoder bandwidth. Linearity therefore grades continuously, blind at one extreme and uniform at the other.

Scene-invariance, by contrast, is binary — which we did not expect. Leave-one-scene-out cross-validation (Figure 3b, x-axis) tests whether a single linear axis transfers across held-out rooms; cross-condition memory-transplant cost (y-axis) tests whether one condition's memory drives another's policy. Blind sits alone in the scene-invariant + transplant-robust quadrant (LOSO $R^2 \approx 0.85$, transplant cost ≈ 0.05). The four sighted conditions — including coarse, despite its high linear readability above — cluster in the scene-conditional + transplant-brittle quadrant. The split sits at the zero-vs-nonzero encoder bandwidth line, not at low-vs-high: any visual input, however coarse, suffices for memory to commit to scene-specific gating. The bottleneck-vs-rich-encoder dichotomy of ğ3.1 is therefore a *magnitude-axis* statement; on the scene-invariance axis the boundary shifts to blind-vs-sighted, which is the structural finding we did not predict.

Subspace identity goes further. The principal angle between every pair of conditions' position-direction subspaces (Figure 3c) puts 9 of 10 cross-condition pairs at $\geq$ 75; only the foveated/foveated-logpolar pair, the two conditions sharing the foveated retina, falls at ~ 67. **The five position codes are otherwise nearly maximally distinct** — five conditions on the same task with the same architecture converge to five almost-disjoint memory subspaces, not the Procrustes-equivalent variants of one shared solution we would naively have expected. The maximal disjointness, surprising on its own, is also the most direct support for the central thesis: sensor structure does not modulate a shared cognitive map, it selects categorically different ones.

The three boundaries do not coincide, and that is the structural finding: linearity grades continuously, scene-invariance flips binary at vision-on/off, subspace identity is condition-specific. The encoder selects a multi-dimensional point in format space, not a value on a single knob. This recasts ğ3.1's magnitude collapse: rich-encoder conditions distribute position into non-linear directions, and what remains linearly-readable is scene-conditional — the two boundaries together, not either alone, produce the policy-readability collapse. Vision is therefore a format lever, not just a magnitude lever, and the only way to a linearly-readable, scene-invariant, transferable cognitive map is to remove vision entirely; the transplant asymmetry on panel (b)'s y-axis [Geva-Sagiv et al., 2016] confirms at the policy-execution level (blind-style memory drives any policy, rich-encoder memory only drives its own). The dichotomy maps onto the no-remapping vs. global-remapping spectrum of hippocampal coding under hidden-state inference [Sanders et al., 2020, Achille and Soatto, 2018], which the agent recapitulates without being trained for it: biologically this predicts blind animals (cave fish, blind mole rats) use a fundamentally different memory format from sighted ones, not a lower-fidelity version of the same code. Manifold-shape scalars and a convergence-time check (Appendix Table 2, Appendix Figure 14c) corroborate the dichotomy at the geometric and developmental levels.

3.3 Temporal Format: blind recomputes the code each step, sighted read it as stationary

The two-route architecture leaves a structural fingerprint in time. **When vision pre-delivers scene structure, memory becomes a passive reader of it from a stable readout axis; when vision is silent, memory has to construct scene structure step by step from motion, and the construction process is non-stationary.** Blind sits at the second extreme, sighted at the first; within sighted, less encoder bandwidth stiffens the readout. The deep claim is not that blind has a cognitive map (Wijmans [Wijmans et al., 2023] showed it does), but that a constructed map and a pre-built map are categorically different objects.

To test which format each condition takes, we train a linear decoder of egocentric goal-vector at trajectory step t_{train} and evaluate it at every other step t_{test} [King and Dehaene, 2014]. If memory holds a pre-built map, a single decoder should transfer across all steps — a square block in the $(t_{\text{train}}, t_{\text{test}})$ heatmap (Figure 4a). If memory builds the map step by step, the readout axis rotates with each motion update and the decoder only works near where it was fit — a diagonal stripe. Blind shows the diagonal; coarse the largest block; fov-LP, foveated, and uniform are graded between. Our pre-registered prediction had the opposite direction (we expected memory-is-the-map to look stationary); the inversion is the substance of the section — a constructed map cannot be stationary, because the construction process is the rotation.

We then ground the dichotomy in two views of memory dynamics. Figure 4(b) plots cumulative L2 path length of $\mathbf{h}_2$ vs step-in-episode (mean $\pm$ 25–75 percentile, all evaluation episodes per condition): blind's curve sits highest at every step — every motion step adds map-construction work — while the four sighted conditions cluster lower and closer together. Figure 4(c) collapses each TGM heatmap of panel (a) to a curve of mean decoder R^2 at trajectory-step lag $|i - j|$, and surfaces the within-sighted spectrum: the decoder-decay ranking inverts encoder bandwidth (Figure 1's graded scale). Panel (b) shows the blind-vs-sighted dichotomy in memory-work terms; panel (c) resolves the within-sighted bandwidth gradient in axis-stationarity terms — a stable axis can carry a changing projected value, so the two panels read complementary aspects.

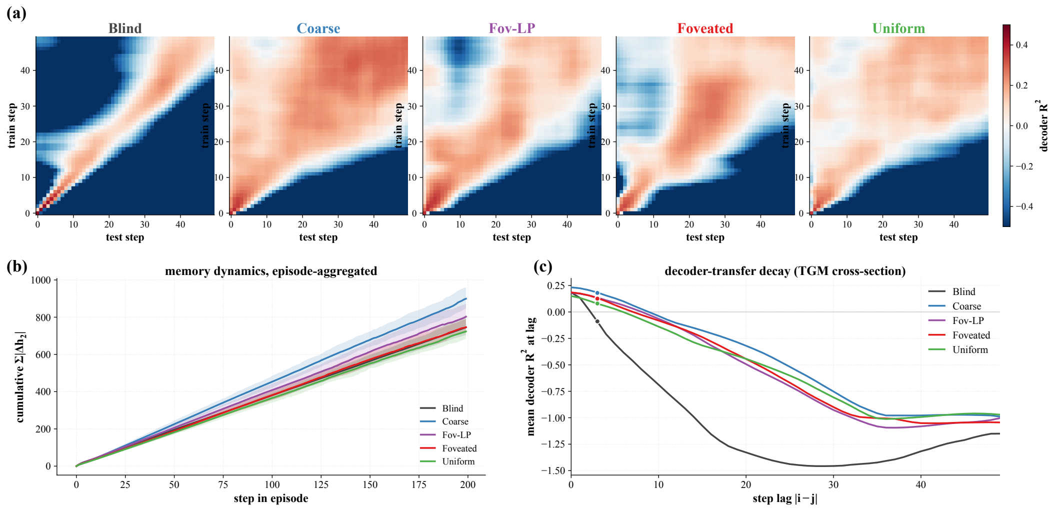


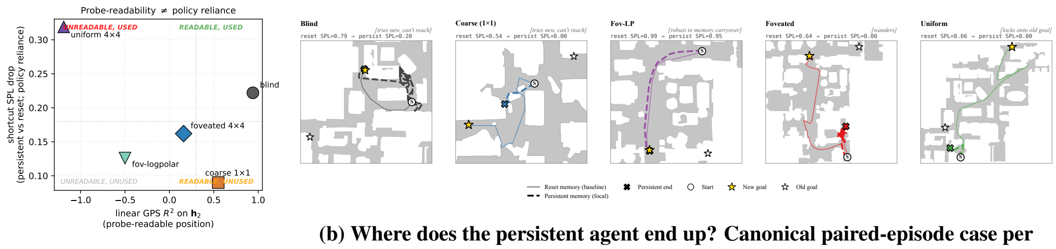
Figure 4: **Temporal format: blind rotates the readout axis step by step; sighted hold it stationary, with stiffness graded by encoder bandwidth.** **(a)** Step-by-step decoder generalisation [King and Dehaene, 2014]: each 50×50 heatmap is decoder R^2 trained at row step, tested at column step. Diagonal-only $\Rightarrow$ rotating basis (blind); square block $\Rightarrow$ stationary readout (coarse); intermediates $\Rightarrow$ fov-LP, foveated, uniform. **(b)** Cumulative memory displacement $\Sigma|\Delta \mathbf{h}_2|$ vs step-in-episode, mean and 25–75 percentile band aggregated over all evaluation episodes per condition: total memory work per step. **(c)** Mean decoder R^2 vs step lag $|i - j|$, each panel-(a) TGM collapsed to a curve: format stationarity. Panel (c) ranks coarse $+0.18 >$ fov-LP/foveated $+0.13 >$ uniform $+0.08$, the inverse of encoder bandwidth (less bandwidth, more stationary). Murray τ and persistent-homology corroborations in App F.

The pattern carries weight beyond the cross-condition contrast. Blind exhibits a slow-prefrontal-vs-fast-sensory cortical-hierarchy timescale gradient [Murray et al., 2014] within a single recurrent network, split by route (encoder fast, integration slow) rather than by anatomical area — route segregation by itself is sufficient for timescale segregation, a structural prediction not yet cleanly testable in biology. Wijmans's "maps in blind agents" [Wijmans et al., 2023] should accordingly not be read as evidence that blind and sighted build the same object: blind builds a step-by-step rotating frame, sighted a world-anchored stationary one, and these afford different downstream uses (planning, transfer, robustness). Stationarity has a structural floor set by the encoder — scaling the recurrent network cannot produce it, because integrating motion from scratch *is* the rotation. The cognitive-map design space is therefore not one knob (capacity) but two: an encoder rich enough to pre-build scene structure, and a recurrent state large enough to integrate it. Murray autocorrelation and persistent-homology corroborations [Gardner et al., 2022] are in Appendix F.

3.4 Consumption: causal intervention dissociates probe-readability from policy reliance

A linear probe tells us *where in memory the position code lives*, but not whether the policy actually uses it. The two could come apart in either direction: a condition might carry a probe-readable position code that the policy ignores ("readable but unused"), or lack one yet behave as if it had one ("unused but readable via a substitute"). Both cases would warn against reading probe results as direct policy mechanism. To test policy reliance, we use a paired same-scene different-goal shortcut protocol [Tolman, 1948]: re-init the recurrent memory between episodes vs. persist it, and read the SPL drop. **Probe-readability and policy reliance dissociate in two off-diagonal cases** — the dissociation is exactly why the multi-method-convergence design we adopt across ğ3 matters: probe-readability is necessary but not sufficient evidence of policy mechanism.

Both off-diagonal cases were unexpected. Coarse carries a linear position code yet has a small shortcut SPL drop (Figure 5a, "readable but unused" — the policy under-weights its own readable code); uniform has *no* linear position code yet has the largest shortcut SPL drop ("unused but readable via a substitute" — the policy reads something else from memory). A direct readout test names the



(a) The dissociation

(b) Where does the persistent agent end up? Canonical paired-episode case per condition

Figure 5: **Probe-readable position is not what drives behaviour.** **(a)** Per-condition shortcut SPL drop (y-axis = policy reliance on persistent memory) vs. linear position decoding from memory (x-axis = probe-readability). Coarse sits in "readable but unused"; uniform in "unused but readable via a substitute" — the two off-diagonal cases. **(b)** Canonical paired-episode case per condition (same scene, two goals; reset = solid line, persistent = dashed bold line, persistent endpoint marked $\times$; new goal = filled star, old goal = open star). Blind and Coarse try the new-goal direction but fail to reach (*tries-new*); Fov-LP successfully reaches the new goal even with persistent memory (*robust to memory carryover*); Foveated wanders; **only Uniform's persistent endpoint sits next to the OLD goal —** the visual / scene-history anchor reading ("head to where I was going last time I was here"). Aggregate margin (avg dist persistent end→new − persistent end→old, over same-floor failures): blind, coarse, foveated, fov-LP all negative (closer to NEW); only uniform positive at $+1.06$ m (closer to OLD, n=38). Full 4×4 failure-mode catalog and boundary-check interventions in Appendix B.

substitute: a scene-id classifier separates scenes substantially better in uniform than in blind, while within-scene linear position decoding goes the other way — uniform's memory state encodes scene identity more strongly than position, the inverse of blind. The qualitative trajectory overlay (Figure 5b) gives the candidate mechanism: re-running each condition on a same-scene different-goal pair, only uniform's persistent agent locks onto the *previous* episode's goal (margin $+1.83$ m closer to old than new); blind, coarse, and foveated still try to reach the new goal (margins -0.38, -0.57, -0.59 m). The simplest reading is a visual / scene-history anchor: "head to where I was going last time I was in this scene".

Two further interventions characterise the consumption mechanism but without separating conditions (Appendix Figure 25). *Memory-init transplant* (re-initialise memory from the previous-episode end-state rather than from zero) drops SPL by a comparable amount across all five conditions: **memory in this architecture is an *activity-slot* store, not a synaptic store [Dorrell et al., 2024]** — persistent memory is bound to the trajectory dynamics, not to the underlying weights. *GPS+compass ablation* from step 0 collapses success to near-zero across all five conditions, confirming the position sensor as the upstream causal source whose integration memory encodes. Classifier protocol, the full 4×4 failure-mode catalog spanning the per-condition margin distribution, and the failed mid-rollout visual-ablation control are in Appendix B.

4 Discussion

4.1 Synthesis: three views of one capacity allocation

A reader could imagine three independent things going on across our five conditions: they differ in how much position lives in memory, how that position is encoded, and whether the policy uses it. We read the three as views of one allocation. The visual encoder and the recurrent integration of the position sensor compete for finite memory capacity, and the encoder's bandwidth tilts the balance: **bottleneck conditions retain a linear position code in memory, rich-encoder conditions reallocate to the visual route, and the policy reads from whichever route the memory makes available.** Each condition therefore occupies a distinct (X, Y, size) coordinate in Figure 6, mirroring the three results sections (magnitude X, format isolation Y, consumption marker-size). Blind sits alone in the linearly-readable + format-isolated quadrant with the largest behavioural reliance; uniform, foveated, and fov-LP cluster in the non-linear-only + format-shared quadrant; coarse falls in the linearly-readable + format-shared quadrant, the off-diagonal location that motivates ğ3.4's

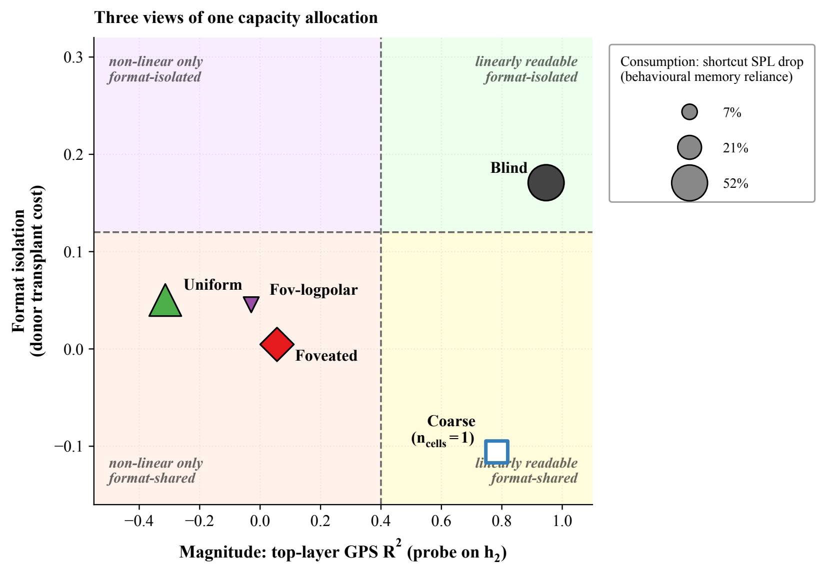


Figure 6: **Three views of one capacity allocation, one point per condition.** X axis: magnitude (top-layer GPS R^2 on $\mathbf{h}_2$, ğ3.1). Y axis: format isolation (donor transplant cost, ğ3.4; higher Y = donor memory is more disruptive to other conditions' policies). Marker size: consumption (shortcut SPL drop, ğ3.4; larger = more policy reliance on persistent memory). Coarse's Y has only $n_{\text{cells}}=1$ (single donor cell available, hollow marker).

dissociation. Coarse and uniform sit on opposite sides of the magnitude axis for different reasons: coarse's encoder cannot supply spatially-resolved features (so capacity defaults to the integration route), while uniform's encoder supplies rich features that capture per-step position (so capacity is reallocated to the visual route, leaving integration unattended). The two are different regimes of one allocation, not the same mechanism.

4.2 Biological precedent and predictions

The hypothesis "sensor niche shapes spatial memory" has long-standing comparative-cognition precedent — rodent dark-rearing [Chen et al., 2016], bat cross-modality remapping [Geva-Sagiv et al., 2016], congenitally blind humans with larger anterior hippocampal volume [Fortin et al., 2008]. We do not claim our silicon results *demonstrate* the bio–AI parallel; we offer them as the patterns one would expect if the hypothesis held in animals, observed in a controlled within-architecture comparison animals do not afford. Loosely mapped: blind $\leftrightarrow$ subterranean rodents [Kimchi et al., 2004]; coarse $\leftrightarrow$ single-channel sensory summaries (echolocation, infrared pit-detection); foveated $\leftrightarrow$ primate / raptor (gaze-fixed); uniform $\leftrightarrow$ insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020].

Two of our findings land cleanly under the hidden-state-inference framework of Sanders et al. [2020], in which place fields remap when sensory evidence supports distinct hidden states. The LOSO catastrophe in rich-encoder conditions matches global-remapping — rich per-step visual evidence supports scene-specific hidden states, and the linear position direction reorganises across rooms; bottleneck conditions exhibit no remapping. Blind's transient-code temporal-generalisation signature and longer intrinsic timescale [King and Dehaene, 2014, Murray et al., 2014] reproduce the cortical-hierarchy slow-integrator pole. Both are "the same metric one would run on monkey or human cortex, applied unchanged to a recurrent navigator". Three predictions follow that are falsifiable in animals as stated. Within-species manipulations of effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-/global-remapping spectrum. Gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells (Wirth et al. 2017 is the closest existing dataset). And the Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings, catalogued separately in cognitive-map theory [Tolman, 1948, O'Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], may all be views of one mechanism. Whether biological hippocampi obey a finite-capacity allocation principle of this kind is an open empirical question.

4.3 Implications and scope

The principle has a simple geometric reading. A finite memory has a fixed number of dimensions to spend; how it spends them depends on what its input already provides. When the encoder delivers rich, near-complete spatial information at every step, the memory has no reason to maintain a separate linearly-readable position code, and the policy is free to commit those dimensions to other content. When the encoder delivers almost nothing, the memory must reconstruct position by integrating the action stream over time, and the resulting code is forced into a form the linear action head can consume. **This is the empirical pattern of ğ3.1–ğ3.2, and it is what the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018] predicts:** bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure 3b). The framing has a quantitative skeleton: a hidden state with effective rank d allocates dimensions among (i) directly carrying encoder-derived features (capacity d_e), (ii) integrating sensor history (capacity d_i), and (iii) carrying task-relevant non-position content (capacity d_o), subject to $d_e + d_i + d_o \lesssim d$. Bottleneck encoders force $d_e \to 0$, leaving d_i to dominate the linear position code; rich encoders enlarge d_e, so policy gradient minimises d_i down to whatever residual the integration route still needs. Our Procrustes / participation-ratio / eigenspectrum-α measurements collectively probe d (effective rank ≈ 4 for blind, lower for coarse where the encoder collapse shrinks d overall, Appendix E.3); the magnitude collapse follows from d_i shrinking as d_e grows. We use IB descriptively rather than mechanistically; Saxe et al. [2019]'s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, non-linear-probe recovery, mutual-information rank in Appendix D.1) is robust to those concerns.

The principle is also architecture-agnostic: a content-level allocation should appear in any system where a finite-capacity memory reads a pretrained encoder. Replacing recurrent gating with attention-over-history should not change the headline ordering. In transformer-based navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022 for the transformer–hippocampus correspondence), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the same monotonic anti-correlation reported in Figure 2a: **attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders**. The cleanest falsification target is direct — train a transformer navigator under our five sensor conditions; the opposite pattern would falsify architecture-independence and bound the principle to recurrent-state systems. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity *away from* the downstream substrate's spatial code, rather than merely fail to teach it, and bandwidth-restricting perception (foveation, log-polar) may be *cognition-enabling* rather than only compute-saving. The relevant variable is encoder spatial-feature *variety* per step rather than raw input bandwidth (Appendix I); a finer multi-point sweep within recurrent agents is left to future work.

As a partial test of architecture-independence, we ported the encoder-bandwidth design to a different environment, encoder family, and training regime (Figure 7). Frozen DINOv2-Base [Oquab et al., 2024] patch features at five resolutions — mirroring our Habitat chassis (blind, coarse, foveated, uniform, foveated-logpolar) — feed a small recurrent memory trained on Memory Maze [Pasukonis et al., 2023] 9×9 offline trajectories with next-feature prediction; we then probe `agent_pos` from the memory state (Ridge linear and 4-layer non-linear probe, frame-level $80/20$ split, eval window steps 250–500). Three pre-registered patterns hold. Linear R^2 rises monotonically with encoder bandwidth (blind 0.01, coarse 0.38, foveated 0.41, fov-LP 0.41, uniform 0.45); non-linear R^2 saturates near 0.998 for all sighted conditions and is at chance for blind (-0.0003); the non-linear gap (non-linear minus linear) decreases monotonically with bandwidth (coarse $+0.62$, foveated $+0.59$, fov-LP $+0.59$, uniform $+0.55$). **Format-shift recovery is largest where linear readability is smallest — the same dichotomy our Habitat result reports, in a different environment, encoder family, and policy regime.** The Memory-Maze setup deliberately omits the position-sensor channel our Habitat agents receive, so the encoder is the only sensory route to absolute position and blind has no signal at all. The bandwidth-allocation *tradeoff* (where blind's integration route compensates) thus requires a competing route; the format-shift dynamic (linearly readable code emerges

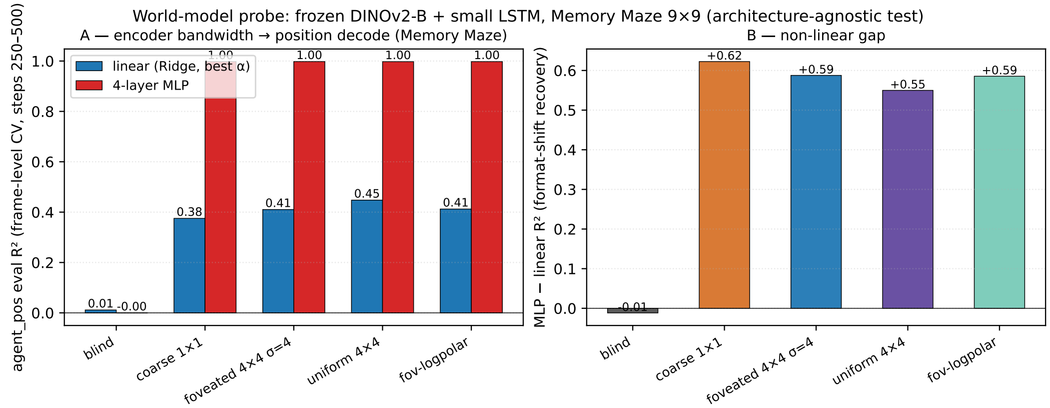


Figure 7: **Architecture-agnostic confirmation: format shift reproduces on Memory Maze with a frozen DINOv2-Base encoder feeding a small LSTM.** *(A)* Linear (Ridge, best of $\alpha \in \{0.1, 1, 10, 100, 1000\}$, blue) and 4-layer MLP (red) eval R^2 for `agent_pos` from the LSTM hidden state, frame-level random CV (80/20), eval window steps 250–500 of T=500 Memory-Maze trajectories. Linear R^2 is monotone-increasing with encoder bandwidth (BLIND $0.01 \rightarrow$ UNIFORM 0.45); MLP R^2 saturates at ≈ 0.998 for all sighted conditions and is at chance for BLIND. *(B)* The non-linear gap (MLP−linear) is monotone-*decreasing* with bandwidth: format-shift recovery is largest where linear readability is smallest (COARSE $+0.62$ vs. UNIFORM $+0.55$), the same dichotomy our Habitat agent shows in Figure 3a. The opposite linear-R^2 direction relative to Habitat reflects the absence of GPS/compass channels in the Memory-Maze setup — without an alternative positional anchor, the encoder is the only route, and BLIND carries no signal.

from a non-linear substrate as bandwidth grows) generalises across both regimes. The two regimes together place the broad encoder-structure claim on a second environment / encoder / policy and clarify that the specific direction (high-bandwidth-rich vs. bottleneck-rich) is mediated by which sensory channels carry positional information.

4.4 Limitations

Single seed per condition is the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]. Cross-condition rigour comes from convergent multi-method evidence: linear and non-linear probes, CKA, Procrustes, subspace divergence, LOSO, lag-k, 5×5 memory transplant, Skaggs spatial information, and excursion-forgetting all rank the conditions in the same direction — unlikely under single-seed noise alone — supplemented by an independent-retraining replication of blind (Appendix A). Per-condition seed-variance error bars over a multi-seed grid are left to follow-up.

Methods we did not report. Dynamical-systems analysis (fixed-point / slow-point structure [Sussillo and Barak, 2013]; Dynamical Similarity Analysis [Ostrow et al., 2023] as a dynamics-aware Procrustes alternative) would speak to format at the level of computational primitives. We piloted DSA on memory-state time-series (10-d PCA pre-reduction, 50-step trajectories) and found the cross-condition signal sensitive to hyperparameter choices, rollout protocol, and episode-phase alignment (blind's median episode length is $\sim 2.5\times$ sighted, so first-T-step samples occupy different behavioural phases); under the cleanest protocol (deterministic-rollout, T=50) sighted clustered at ~ 0.10 angular distance while blind sat further out at ~ 0.27, but blind's within-condition split-half was also ~ 0.23, leaving the signal-to-noise ratio insufficient. Information-theoretic estimators of $I(h_t; \mathrm{pos}_{t+k})$, gaze-placement controls, and a $\sigma_{\max}$ strength sweep within the foveated condition are similarly left to follow-up.

Architecture, algorithm, and task scope. We use a recurrent LSTM with DD-PPO, PointGoal, and a ResNet-18 visual encoder, matching the canonical pipeline of Wijmans et al. [2023], Banino et al. [2018], Cueva and Wei [2018] for direct comparability. The architecture choice is replication-driven; the principle itself is stated at the encoder–memory interface and is architecture-agnostic in principle. A reasonable concern is whether the encoder–memory race is an LSTM gating artefact, but two within-paper checks argue against it: non-linear (2-layer MLP) probes recover position in rich-encoder conditions ($R^2 \in [0.35, 0.79]$), so the memory does not discard position — only the linear-readable form tracks encoder bandwidth, a property of the linear policy head; and the cross-condition memory transplant (ğ3.4) is a behavioural test outside any probe framework and reproduces the same ordering. The cleanest falsification target is a transformer-based navigator under the same conditions, identified in ğ4 but not verified here; the magnitude and format patterns could also shift under different RL algorithms, navigation tasks (ObjectGoal, ImageGoal, exploration), or visual encoders.

5 Conclusion

A five-condition visual-sensor ablation in a recurrent PointNav agent shows sensor structure as a content variable whose effects cohere under a single capacity-allocation principle, read along three views: magnitude (the bandwidth–allocation tradeoff between encoder and recurrent-integration routes), format (within-condition scene-invariance, across-condition non-interchangeable subspaces with asymmetric memory transplant, and a King–Dehaene-style temporal-format dissociation in which blind's code rotates while sighted readouts stay stationary), and consumption (probe-readability dissociates from policy reliance, with the off-diagonal cases naming a scene-history substitute the rich encoder reads instead of position). Several findings inverted the predictions our framework would naively have suggested: the bandwidth axis tracks per-step feature variety rather than encoder cell count, scene-invariance flips binary at vision-on/off rather than grading smoothly, blind's memory rotates rather than stabilises (because constructing a map step-by-step *is* the rotation), and the five conditions converge to nearly-orthogonal subspaces rather than Procrustes-equivalent variants of one solution — divergences we treat as load-bearing rather than as exceptions, since each sharpens the framework into a more falsifiable claim. The bandwidth ordering is preserved on held-out MP3D and reproduces in a different encoder family / environment / policy regime (Memory Maze with frozen DINOv2-Base, Figure 7), and the pattern is consistent at the high level with hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-/global-remapping axis under hidden-state inference [Sanders et al., 2020] — offered as an open bio–AI hypothesis the comparative-cognition literature can test directly. The encoder–memory race is stated at the interface level and should in principle generalise to transformer-based navigators, the cleanest falsification target. Methodologically, we view the paper as a step toward a comparative cognitive neuroscience of artificial navigation agents: matched-condition populations of trained agents analysed with the same metrics (population geometry, intrinsic timescales, persistent homology, transplant cost, dynamical similarity) used to map biological cortex and hippocampus. The goal is not that artificial agents *are* brains, but that the analytical machinery transfers, the relevant variables can be held fixed, and the cross-condition contrasts yield theoretically informative signatures of the same kind comparative neuroscience uses to understand sensor-niche adaptation in animals.

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: ğ3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in ğ3.1.

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   Justification: ğ3.1 specifies training pool and eval set; ğ3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, HewittLiang control). Appendix A gives the ablation's loss formulation and coefficient.

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   Justification: ğ3.7 reports per-condition training budget (57 days on 2ŒV100 for 250M frames) and probing budget (<1h per condition on one V100).

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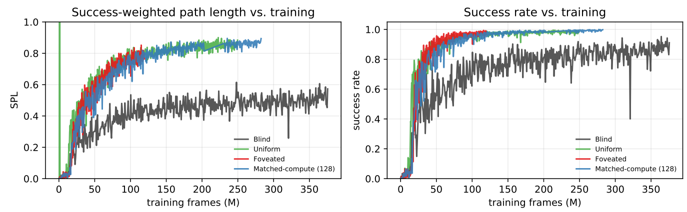


Figure 8: **Behavioural convergence (DD-PPO training curves).** TensorBoard event-file scalars across the 250M-frame training budget. All four sighted conditions (COARSE, FOVEATED, FOVEATED-LOGPOLAR, UNIFORM) converge to SPL 0.82–0.85 and success $> 97\%$ by ~ 100M frames; BLIND converges to SPL ~ 0.50 over a longer horizon. Probing windows fall well after convergence for all conditions, so the encoded-content gap reported in ğ3 cannot be attributed to differential under-training.

A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (a per-channel running normalisation layer applied to the visual stream before the ResNet-18 backbone) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected by this choice; an ablation that re-enables the normaliser (with the autograd conflict patched) is available for future work that wants to revisit it.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) a pre-update gradient sanitisation pass replaces NaN/Inf gradients with zero before each PPO step; (ii) a forward-looking geodesic-distance clamp prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories and apply identically to all five conditions.

Log-polar foveation transform The log-polar control (Appendix I) applies a differentiable log-polar resampling step before the ResNet-18 backbone: the 128×128 avg-pooled image is mapped onto a 64×64 grid with n_ρ=64 log-spaced radial samples and n_θ=64 angular samples, mimicking the cortical magnification of mammalian primary visual cortex. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation Under deterministic rollouts, episode lengths span $\sim$100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the magnitude ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

LSTM Layer 0 sensor branch The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely. The full per-layer breakdown (linear R^2 at $\mathbf{h}_0, \mathbf{h}_1, \mathbf{h}_2$ and at the cell states $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2$) is in Figure 10: Layer 0 carries GPS for every condition; Layer 1 dips for everyone; Layer 2 separates blind/coarse from foveated/uniform; cell states do not carry the GPS code as cleanly as hidden states. The headline "top-layer linearly-decodable GPS" claim is therefore specifically about $\mathbf{h}_2$, not the cell pathway.

Non-linear (MLP) probe details For each condition we fit a 2-layer MLP (hidden $256{-}128$, ReLU, L_2 weight decay 10^{-4}, sklearn `MLPRegressor` with early-stopping) on the same top-layer hidden states and GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. The full probe-depth sweep (linear → MLP-1 → MLP-2 → MLP-4) shown in Figure 3a establishes a depth-graded recovery: bottleneck conditions plateau near $R^2 \approx 0.95$ from depth-0 (linear suffices), while rich-encoder conditions ramp up sharply with probe depth. Top-layer MLP-2 recovery (Ridge linear → MLP-2): blind $0.92 \rightarrow 0.98$, coarse $0.51 \rightarrow 0.75$, FOVEATED $0.06 \rightarrow 0.79$, FOVEATED-LOGPOLAR $-0.07 \rightarrow 0.63$, UNIFORM $-2.28 \rightarrow 0.35$. The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix I) targets that question at one intermediate point.

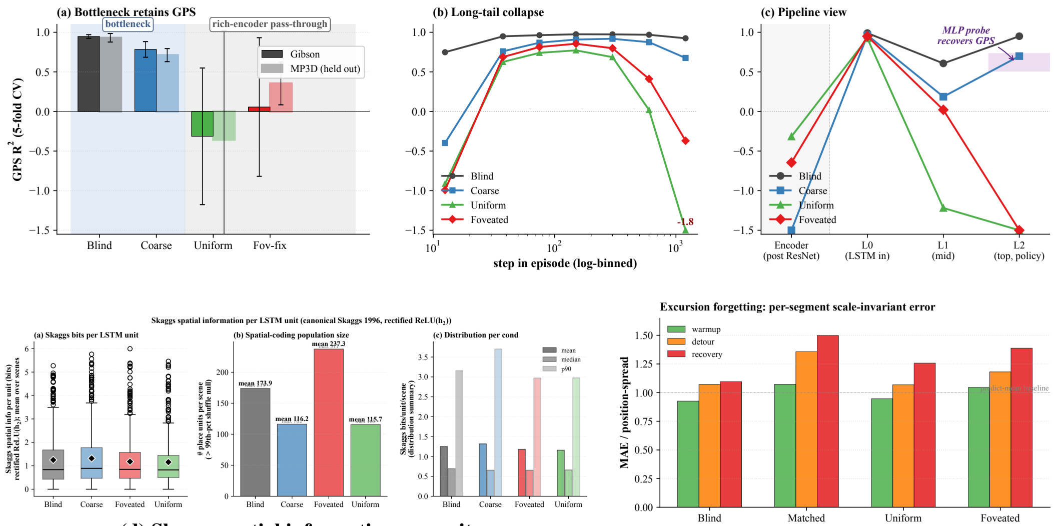
C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); training curves (Figure 8) confirm that all five conditions converge behaviourally before the probing window. The encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. A direct test: distance-to-goal (DtG) decodes robustly across all four conditions ($R^2 \geq 0.81$; Figure 17) while the goal-direction component is at chance everywhere (PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it). The linear probe is therefore capable of fitting position-correlated structure when it exists; the chance-level rich-encoder GPS R^2 is a hidden-state property, not a probe artefact. The real-vs-permuted-label gap (Hewitt–Liang selectivity) likewise tracks GPS R^2.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Three complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear (MLP) probe:* a 2-layer MLP recovers $R^2 \in [0.35, 0.79]$ from rich-encoder top-layer states (Appendix B). *(c) Non-linear similarity (t-SNE).* Beyond linear similarity (CKA $< 10^{-4}$ off-diagonal, Figure 15), a non-linear t-SNE embedding (Figure 18) also separates the four conditions with 1-NN purity 1.000 vs. chance 0.20 — the format divergence is not an artefact of linear-similarity measures. H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.



(d) Skaggs spatial information per unit

(e) Excursion forgetting

Figure 9: **Magnitude: corroborating diagnostics (referenced from ğ3.1 and Table 1).** *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; the MP3D shift is single-seed and awaits replication). *(b)* GPS R^2 binned by step-in-episode (7 log-spaced bins): rich-encoder readability holds in the mid-episode window then collapses in the long tail. *(c)* Pipeline view: GPS R^2 along Encoder $\to$ L0 (LSTM input) $\to$ L1 $\to$ L2 (single-fit; the 5-fold CV version is condensed in main Figure 2). Readability is comparable across conditions at the LSTM input and only diverges at L2; the shaded band at L2 is the 2-layer MLP probe recovery range. *(d)* Per-unit Skaggs spatial information [Skaggs et al., 1996] is condition-flat at the headline metric; the per-unit regression analysis in ğ3.1 (trajectory-phase vs. raw-position R^2) is the more diagnostic reading. *(e)* Mid-episode excursion forgetting (25-step random-action perturbation; per-segment scale-invariant MAE): blind fragments most, consistent with the strongest reliance on the integrated action stream.

D Magnitude: corroborating diagnostics

The main-text Figure 2 (ğ3.1) summarises the magnitude story across three views: cross-condition R^2, training-trajectory dynamics, and predictive horizon. This appendix collects the corroborating diagnostics referenced from ğ3.1: the composite of cross-environment generalisation / step-binned readability / pipeline view / Skaggs information / excursion forgetting (Figure 9); the per-LSTM-layer breakdown of where the GPS code lives (Figure 10); a single-fit companion check (Figure 11); the GPS+compass version of the substitution-dynamics curves (Figure 12); and the nat-scale information-theoretic capacity estimate (Figure 13, ğD.1).

D.1 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in ğ3.1 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \text{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X;Z) \geq \sup_{T} \mathbb{E}_{P_{XZ}}[T(x,z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x,z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \text{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

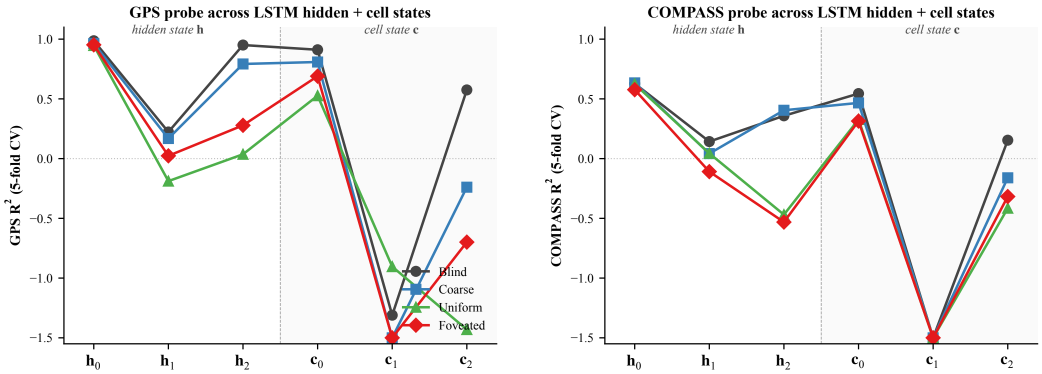
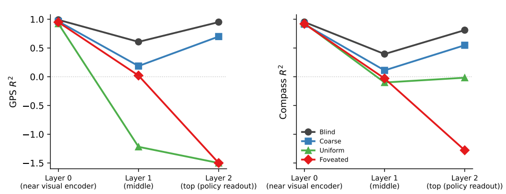
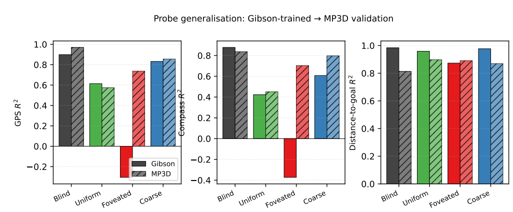


Figure 10: **Where GPS lives along the LSTM stack: per-layer linear probe R^2.** 5-fold CV linear probe at each hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ — the GPS sensor embedding enters the LSTM concat at Layer 0 regardless of vision condition (Appendix B). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through the middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for the bottleneck conditions. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The headline "top-layer linearly-decodable GPS" result in ğ3.1 is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 9a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-rich-encoder partition is the same.

Figure 11: **Single-fit per-layer and MP3D companion checks.** Companion to the headline 5-fold CV results: a single train/test split reproduces the bottleneck-vs-rich-encoder partition reported in Table 1, ruling out CV-fold-induced artefacts. The MP3D panel re-evaluates the same checkpoints on a held-out scene distribution; the partition still holds.

E Spatial Format: corroborating diagnostics

The main-text Figure 3 (ğ3.2) shows three convergent measures of format divergence. This appendix collects: the full transplant matrix together with cross-condition probe transfer and subspace evolution (Figure 14); a linear-similarity null and a non-linear similarity check (Figures 15, 18); the geometric structure of the position code within $\mathbf{h}_2$ (Section E.1); a population-coding analysis (Figure 16); the eigenspectrum power-law per condition with associated geometric scalars (Section E.3); and a causal subspace ablation (Section E.2).

E.1 Capacity-allocation: additional geometric tests

The figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. 3a) and LOSO scene-invariance (Fig. 3b) results in ğ3.1. (The cross-condition probe-transfer matrix is shown as panel (b) of Figure 14; we do not duplicate it here.)

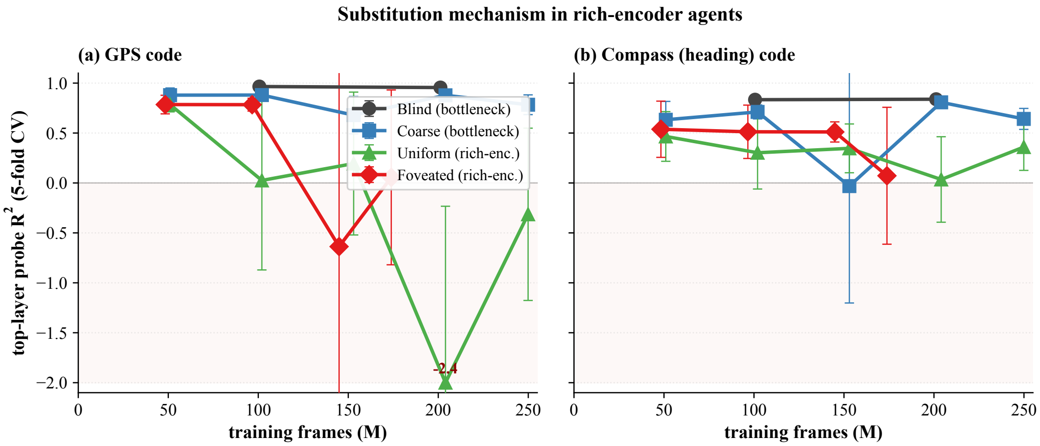


Figure 12: **Cross-training substitution dynamics: GPS and compass (companion to main Figure 2b).** Probe R^2 across training checkpoints, truncated at 250M frames. *(a)* Top-layer GPS R^2 (the source panel for Figure 2b). *(b)* Top-layer compass R^2 tells the same story at smaller magnitude — substitution applies to spatial codes broadly, not just position.

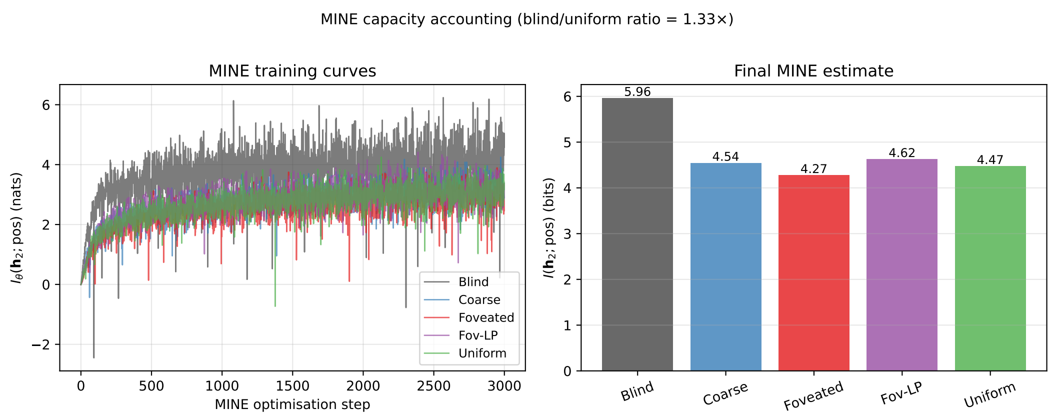
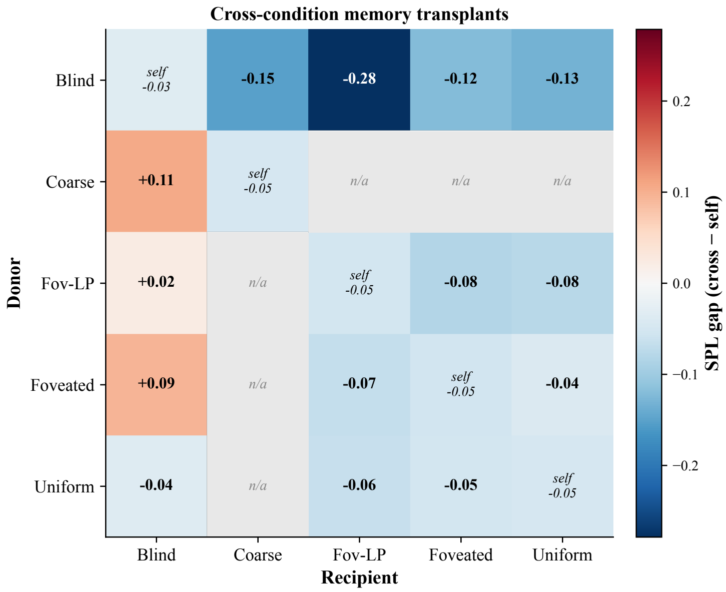


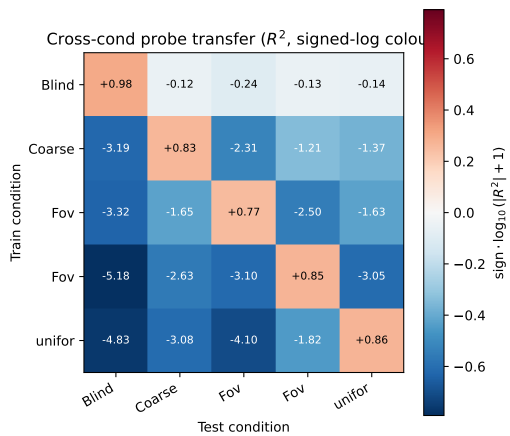
Figure 13: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim$ 3,000 steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06, Coarse 4.55 ± 0.03, Foveated 4.30 ± 0.04, Foveated-logpolar 4.66 ± 0.04, Uniform 4.47 ± 0.06. Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std ≤ 0.06 bits, much smaller than the within-sighted gap ≈ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 3b).

E.2 Single-direction β-scrubbing reveals redundant population coding

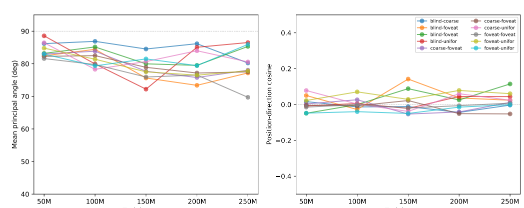
The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).



(a) $5{\times}5$ **memory-transplant matrix (full)**



(b) Cross-condition probe transfer



(c) Subspace evolution over training

Figure 14: **Spatial Format: corroborating diagnostics (referenced from ğ3.2).** *(a)* The full $5{\times}5$ memory-transplant matrix at midpoint $t{=}30$ (the per-recipient mean enters main Figure 3b as the "transplant cost" axis); the matrix shows the same direction-asymmetry condensed there. *(b)* Cross-condition probe transfer matrix (train on row, test on column): off-diagonal R^2 is catastrophically negative ($-3.5{\times}10^2$ to $-3.5{\times}10^4$ unclipped; clipped to $[-1, 1]$ for visualisation) — the position-axes encoded across conditions are jointly incompatible at the readout level, even when the same scenes are visited and the same task is solved. *(c)* Subspace divergence is established early: principal angles are already near-orthogonal at the earliest probed checkpoint ($\sim$ 50M frames) and stay so throughout, indicating the structural separation is laid down well before convergence.

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	$+0.95$	$+0.93$	$+0.94$	$+0.95$
Coarse	$+0.72$	$+0.62$	$+0.79$	$+0.84$
Uniform	-1.08	-1.46	$+0.42$	$+0.31$

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative

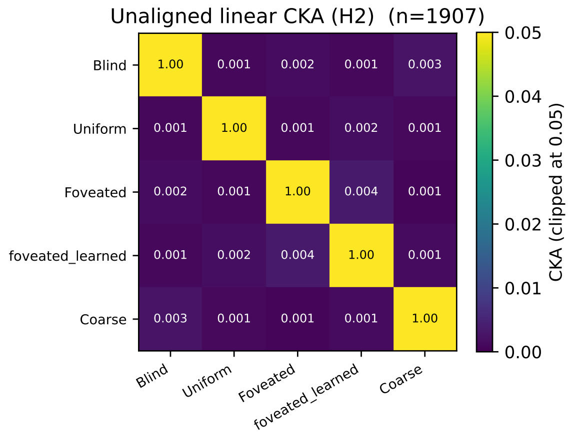


Figure 15: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 14); CKA is the least informative of the three format-divergence measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

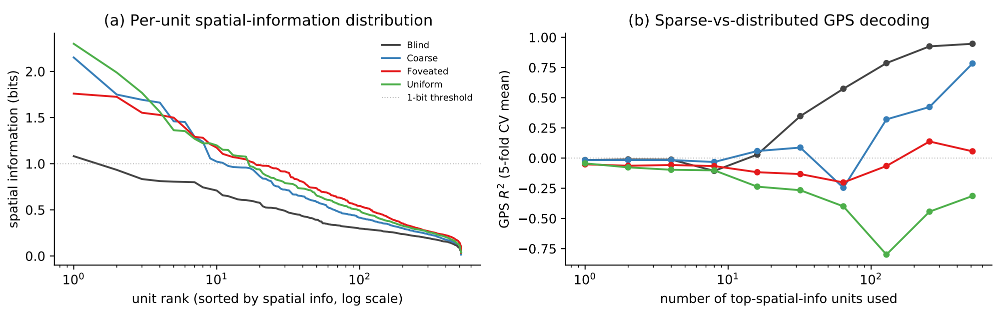


Figure 16: **Population-coding analysis: per-unit spatial information vs. population readout (referenced from ğ3.2)**. *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max $\approx$ 2 bits) than bottleneck conditions; blind has the flattest distribution (max $\approx$ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

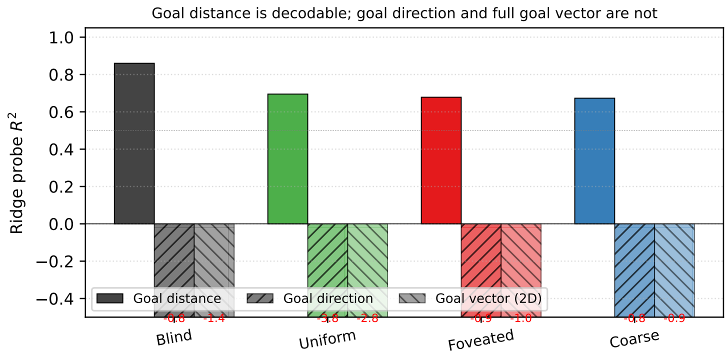


Figure 17: **Goal-vector probe R^2 (referenced from Appendix C, "probe-capacity rebuttal).** Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

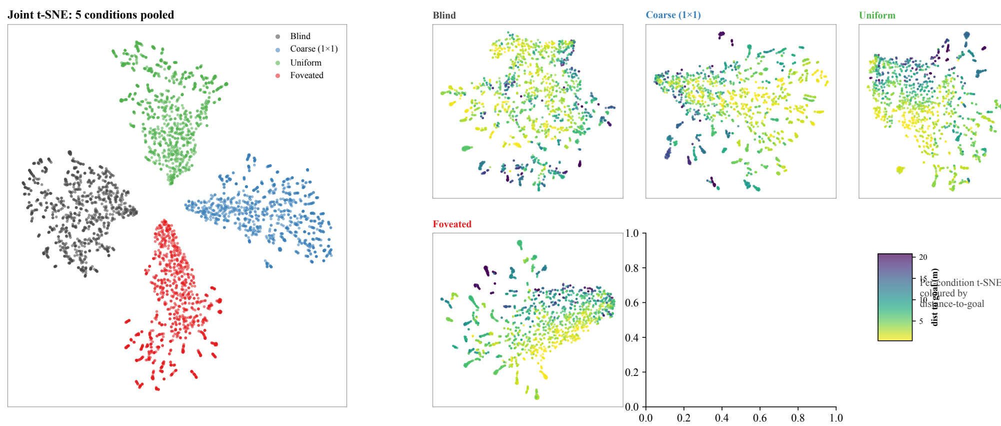


Figure 18: **Non-linear similarity: t-SNE of top-layer hidden states (referenced from ğ3.2).** *Left:* joint t-SNE of 5,000 samples (1,000 per condition) in 512-d hidden space (perplexity 30). The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal (n=2,000 per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with ğ3.1: blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

E.3 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 22, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

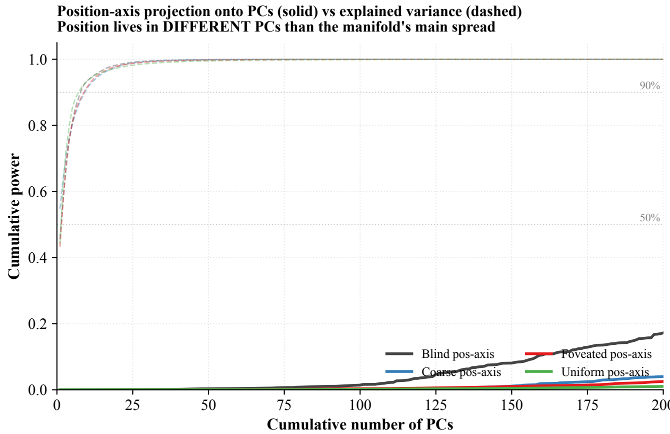


Figure 19: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 3b).

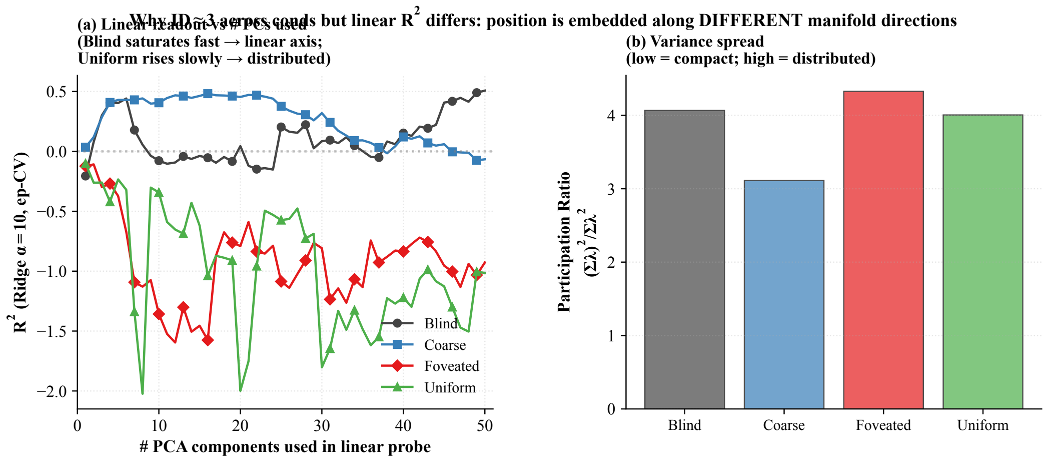


Figure 20: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with $\leq$ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 19).

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading), and far above Stringer's V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind's eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx$ 1.7–1.9, all near the $1 + 2/d_{\text{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim con-

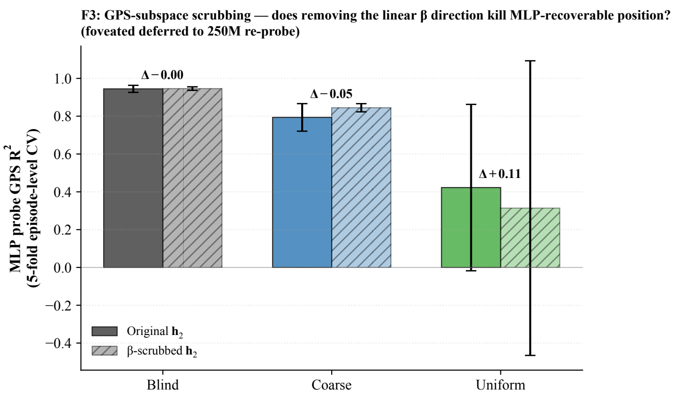


Figure 21: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^+\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

Table 2: **Geometric scalars: per-condition manifold shape (companion to Figure 22; referenced from ğ3.2).** Participation ratio (PR), intrinsic dimension (ID-MLE / ID-CDF), and eigenspectrum power-law exponent α on $\mathbf{h}_2$. All values from post-retrain hp-consistent agents (3-seed mean over 30,000-step subsamples; std < 0.05). Blind separates on α (sharper tail) and on ID (lowest); the four sighted conditions cluster on α and ID (1.73–1.92, 2.04–2.41). PR varies more across sighted conditions and does not track encoder bandwidth monotonically; we read it as different geometric distributions of the visual route's residual variance rather than a clean ordering.

Condition	PR	ID-MLE	ID-CDF	α
Blind	4.41	$\mathbf{1.72}$	$\mathbf{1.54}$	$\mathbf{3.63}$
Coarse	5.25	2.41	2.20	1.81
Foveated-logpolar	1.70	2.25	2.04	1.88
Foveated	9.11	2.41	2.30	1.73
Uniform	3.09	2.32	2.17	1.92

centrated regime, while the four sighted conditions distribute variance across more PCs. The split bottleneck-vs-rich-encoder reading: blind's heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose 1×1 encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

F Temporal Format: corroborating diagnostics

The main-text ğ3.3 reports three views of the temporal-format dichotomy: the temporal-generalisation matrix (heatmaps), cumulative memory displacement, and decoder-transfer decay. This appendix collects three further diagnostics referenced from ğ3.3: a Murray-style per-unit intrinsic-timescale check (Section F.1), a predictive-horizon test grounded in successor-representation theory (Section F.3), and a topological-distinguishability test via persistent homology (Section F.2).

F.1 Murray-style intrinsic timescale (per-unit autocorrelation)

A separate cogneuro-style timescale check follows Murray et al. [2014]: for each unit in $\mathbf{h}_2$, compute the within-episode autocorrelation curve, average across 256 random units $\times$ 80 random episodes per condition, and read τ as the lag at which the mean curve crosses $1/e$. Figure 23 shows that blind units integrate over a $\tau \approx$ 12-step timescale while every sighted condition crosses at $\tau \approx$ 7–8, recapitulating the slow-prefrontal-vs-fast-sensory cortical-hierarchy gradient [Murray et al., 2014] within a single agent — here split by route (encoder fast, integration slow) rather than by anatomical

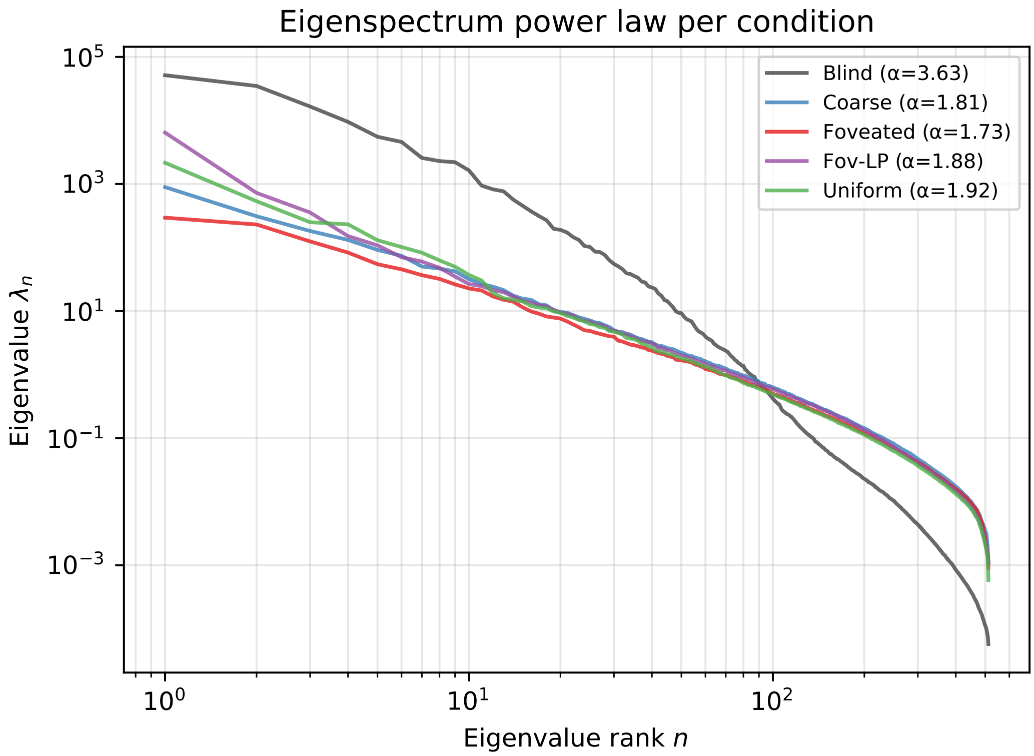


Figure 22: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

area. Within sighted, the four conditions cluster near-identically at the unit-autocorrelation level; the within-sighted bandwidth-inverse stiffness signal lives in the decoder-decay view (main-text Figure 4(c)) rather than in per-unit timescale.

F.2 Topological distinguishability via persistent homology

A parameter-free pipeline used for toroidal grid-cell structure in mouse cortex [Gardner et al., 2022] confirms that conditions are topologically distinguishable: cross-condition Wasserstein-2 distance between H1 persistence diagrams is roughly twice the within-condition seed noise (Figure 24b). Persistent-loop counts order monotonically with encoder bandwidth (blind highest, uniform lowest; Figure 24a). We treat this as suggestive corroboration rather than independent mechanistic evidence. Our pre-registered prediction was the opposite direction (we expected uniform highest, blind lowest); we report what we found.

F.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

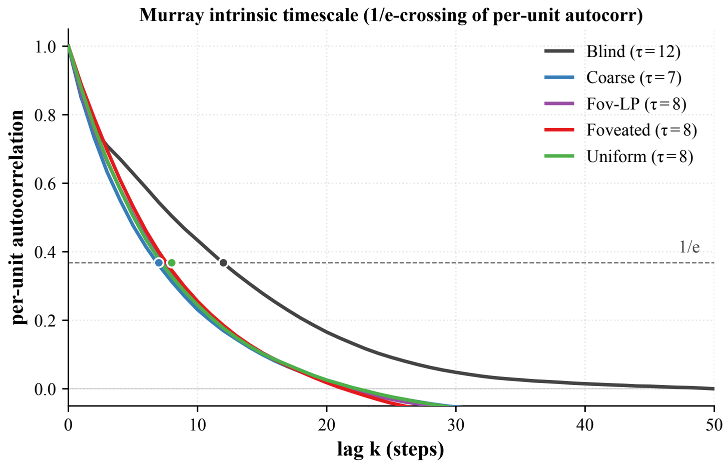
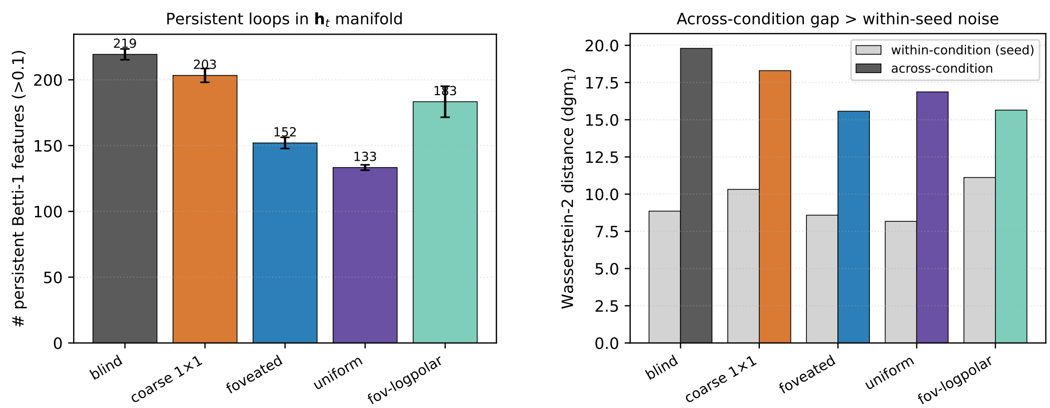


Figure 23: **Murray-style intrinsic timescale per condition.** Per-unit autocorrelation curves averaged over 256 random units $\times$ 80 random episodes per condition; coloured dot marks the lag at which each curve crosses $1/e$ (the intrinsic timescale τ). Blind $\tau \approx 12$; every sighted condition $\tau \approx 7$–8.



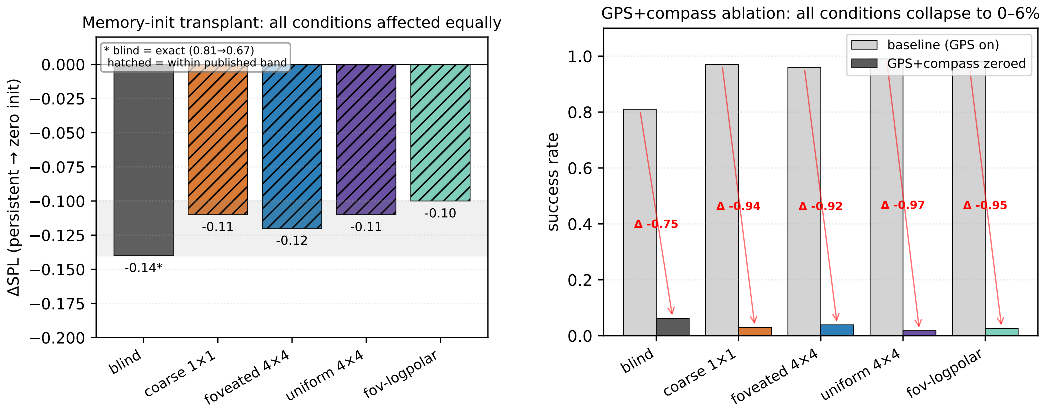
(a) Persistent loops (Gardner 2022)

(b) Across-condition gap > seed noise

Figure 24: **Topological corroboration of the temporal-format dichotomy.** *(a)* Persistent-loop counts (number of H_1 features with persistence > 0.1) per condition. *(b)* Wasserstein-2 distance between H_1 persistence diagrams: the cross-condition gap exceeds within-condition (seed) noise across all five conditions, so the $\mathbf{h}_t$ manifolds are topologically distinguishable.

	$k=0$	$k=1$	$k=5$	$k=10$	$k=20$	$k=50$
Linear (Ridge $\alpha=10$) R^2, all-episode pooling, episode-level 5-fold CV						
Blind	$+0.94$	$+0.94$	$+0.94$	$+0.92$	$+0.90$	$+0.79$
Coarse	$+0.57$	$+0.57$	$+0.57$	$+0.56$	$+0.50$	$+0.37$
Foveated	$+0.06$	$+0.06$	$+0.08$	$+0.13$	$+0.19$	$+0.06$
Foveated-logpolar	$+0.17$	$+0.18$	$+0.17$	$+0.11$	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x,y)_{t+k}$ remains ≥ 0.90 from $k=0$ to $k=20$ (only 0.04-point drop), and 0.79 at $k=50$. The same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2=0.94$ also decodes positions 20 steps in the future at $R^2=0.90$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k=0$–20, dropping to $+0.37$ at $k=50$). *(iii) Rich-*



(a) Memory-init transplant **(b) GPS+compass ablation**

Figure 25: **Boundary-check causal interventions (referenced from ğ3.4).** *(a) Memory-init transplant:* re-initialising memory from the previous episode's end-state (instead of zero) drops SPL by a comparable amount across all five conditions. The cross-condition uniformity is the finding — every agent's policy expects a zero memory state at $t=0$, so memory acts as an activity-slot store bound to the trajectory dynamics. *(b) GPS+compass ablation:* zeroing GPS and compass from step 0 collapses success to near-zero across all conditions. The drops cluster tightly across the four sighted conditions; blind drops less because its baseline is already lower. The position sensor is the upstream causal source whose integration the memory encodes.

encoder conditions have no clear predictive horizon. Foveated stays in $[0.06, 0.19]$ across $k = 0$–50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with the LOSO scene-invariance reading in ğ3.2, but adding the temporal aspect: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

G Consumption: corroborating diagnostics

The main-text ğ3.4 reports the persistent-memory failure-mode test as the condition-discriminating intervention; this appendix presents two boundary-check interventions that characterise the consumption mechanism without separating conditions (Figure 25), the full 4×4 failure-mode catalog spanning the per-condition margin distribution (Figure 26), and a memory-transplant midpoint sweep that defends the $t=30$ choice in the main-text 5×5 transplant matrix (Figure 27).

G.1 Boundary checks: memory-init transplant and GPS-sensor ablation

Memory-init transplant re-initialises the agent's recurrent memory from the previous-episode end-state instead of from zero; SPL drops by a comparable amount across all five conditions (Figure 25a). The cross-condition uniformity is itself the finding: every policy at $t=0$ expects a zero memory state, so memory in this architecture is an *activity-slot* store [Dorrell et al., 2024] bound to the trajectory dynamics, not a synaptic store. *GPS+compass ablation* from step 0 collapses success to near-zero across all conditions (Figure 25b): the position sensor is the upstream causal source whose integration the recurrent memory encodes. The absolute drops cluster tightly across the four sighted conditions and do not cleanly separate them, which is consistent with the upstream sensor being a shared causal bottleneck rather than a discriminator.

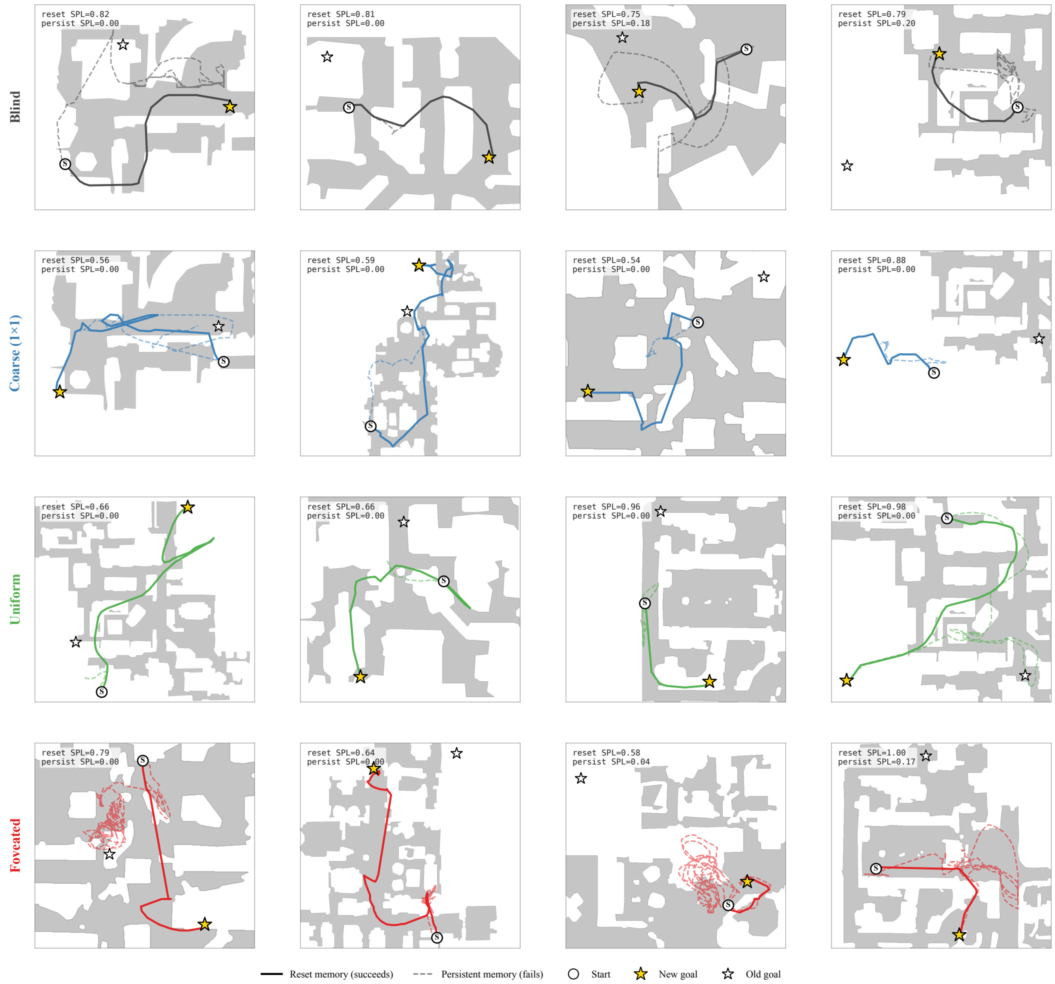


Figure 26: **Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution.** Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see ğ4.4).

G.2 Persistent-memory failure-mode catalog

The main-text Figure 5 shows one canonical paired-episode case per condition; the full $4{\times}4$ catalog below (Figure 26) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$ (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

G.3 Memory-transplant midpoint sweep: defending the $t{=}30$ choice

The main-text $5{\times}5$ memory-transplant matrix is reported at a fixed midpoint $t{=}30$ (the full matrix appears as panel (a) of Figure 14). A natural concern is whether the asymmetry pattern is specific to that midpoint. The midpoint sweep below addresses this: across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps, the bottleneck-vs-rich-encoder gap appears for $t \in$

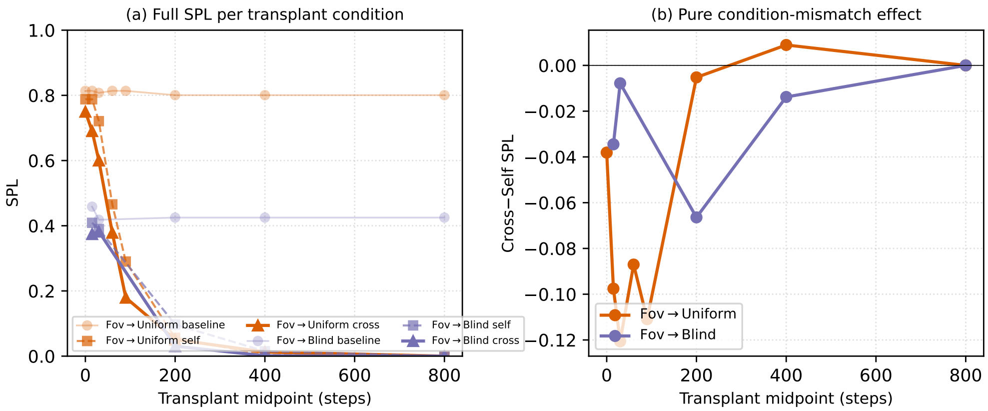


Figure 27: **Memory-transplant midpoint sweep (referenced from ğ3.4).** Two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in main Figure 3 is the t=30 slice. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (the diagonal in the main-text matrix is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see ğ4**

$[30, 90]$ and decays toward zero by $t \geq 400$, with a protocol-noise floor at t=0 where the donor's hidden state has had no time to integrate.

H Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from the main paper. All four are submitted.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

I Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2\times2$, intermediate between coarse's 1×1 and uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another aspect. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, $\sim$ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 2b) corroborates this reading: log-polar starts at $R^2 = +0.92$ at $\sim$ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at $\sim$ 250M frames each) was outside this submission's compute window.